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Cover for fluid systems and related methods

Abstract

Embodiments of a high-pressure, high power, reciprocating positive displacement fluid pumping system and methods are included. The system may include a high-pressure, high power, reciprocating positive displacement pump including a pump plunger, a fluid end block assembly, and a fluid cover. The fluid end block assembly may include a fluid end block body, a suction port, a discharge port, a pump bore positioned in and extending through the fluid end block body, and a fluid chamber positioned in the fluid end block body and in fluid communication with each of the suction port, the discharge port, and the pump bore. The fluid chamber has an open end portion, and the pump plunger may be positioned to move in the pump bore to pressurize one or more fluids in the fluid chamber. The fluid cover includes a monolithic body having a first portion and a second portion, the first portion being received in the open end portion and sealably engaged with the fluid end block body, the second portion being mechanically connected to the fluid end block body.

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10610842	12/2019	Chong	N/A	N/A
10662749	12/2019	Hill et al.	N/A	N/A
10711787	12/2019	Darley	N/A	N/A
10738580	12/2019	Fischer et al.	N/A	N/A
10753153	12/2019	Fischer et al.	N/A	N/A
10753165	12/2019	Fischer et al.	N/A	N/A
10760556	12/2019	Crom et al.	N/A	N/A
10794165	12/2019	Fischer et al.	N/A	N/A
10794166	12/2019	Reckels et al.	N/A	N/A
10801311	12/2019	Cui et al.	N/A	N/A
10815764	12/2019	Yeung et al.	N/A	N/A
10815978	12/2019	Glass	N/A	N/A
10830032	12/2019	Zhang et al.	N/A	N/A
10830225	12/2019	Repaci	N/A	N/A
10859203	12/2019	Cui et al.	N/A	N/A
10864487	12/2019	Han et al.	N/A	N/A
10865624	12/2019	Cui et al.	N/A	N/A
10865631	12/2019	Zhang et al.	N/A	N/A
10870093	12/2019	Zhong et al.	N/A	N/A
10871045	12/2019	Fischer et al.	N/A	N/A
10892596	12/2020	Enya et al.	N/A	N/A

10895202	12/2020	Yeung et al.	N/A	N/A
10900475	12/2020	Weightman et al.	N/A	N/A
10907459	12/2020	Yeung et al.	N/A	N/A
10927774	12/2020	Cai et al.	N/A	N/A
10927802	12/2020	Oehring	N/A	N/A
10954770	12/2020	Yeung et al.	N/A	N/A
10954855	12/2020	Ji et al.	N/A	N/A
10961614	12/2020	Yeung et al.	N/A	N/A
10961908	12/2020	Yeung et al.	N/A	N/A
10961912	12/2020	Yeung et al.	N/A	N/A
10961914	12/2020	Yeung et al.	N/A	N/A
10961993	12/2020	Ji et al.	N/A	N/A
10961995	12/2020	Mayorca	N/A	N/A
10968837	12/2020	Yeung et al.	N/A	N/A
10982523	12/2020	Hill et al.	N/A	N/A
10989019	12/2020	Cai et al.	N/A	N/A
10989180	12/2020	Yeung et al.	N/A	N/A
10995564	12/2020	Miller et al.	N/A	N/A
11002189	12/2020	Yeung et al.	N/A	N/A
11008950	12/2020	Ethier et al.	N/A	N/A
11015423	12/2020	Yeung et al.	N/A	N/A
11015536	12/2020	Yeung et al.	N/A	N/A
11015594	12/2020	Yeung et al.	N/A	N/A
11022526	12/2020	Yeung et al.	N/A	N/A
11028677	12/2020	Yeung et al.	N/A	N/A
11035213	12/2020	Dusterhoft et al.	N/A	N/A
11035214	12/2020	Cui et al.	N/A	N/A
11047379	12/2020	Li et al.	N/A	N/A
11053853	12/2020	Li et al.	N/A	N/A
11060455	12/2020	Yeung et al.	N/A	N/A
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11068455	12/2020	Shabi et al.	N/A	N/A
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11085282	12/2020	Mazrooee et al.	N/A	N/A
11092152	12/2020	Yeung et al.	N/A	N/A
11098651	12/2020	Yeung et al.	N/A	N/A
11105250	12/2020	Zhang et al.	N/A	N/A
11105266	12/2020	Zhou et al.	N/A	N/A
11109508	12/2020	Yeung et al.	N/A	N/A
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11125066	12/2020	Yeung et al.	N/A	N/A
11125156	12/2020	Zhang et al.	N/A	N/A
11129295	12/2020	Yeung et al.	N/A	N/A
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11149533	12/2020	Yeung et al.	N/A	N/A
11149726	12/2020	Yeung et al.	N/A	N/A
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11168681	12/2020	Boguski	N/A	N/A
11174716	12/2020	Yeung et al.	N/A	N/A
11193360	12/2020	Yeung et al.	N/A	N/A
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11220895	12/2021	Yeung et al.	N/A	N/A
11236739	12/2021	Yeung et al.	N/A	N/A
11242737	12/2021	Zhang et al.	N/A	N/A
11243509	12/2021	Cai et al.	N/A	N/A
11251650	12/2021	Liu et al.	N/A	N/A
11261717	12/2021	Yeung et al.	N/A	N/A
11268346	12/2021	Yeung et al.	N/A	N/A
11280266	12/2021	Yeung et al.	N/A	N/A
11306835	12/2021	Dille et al.	N/A	N/A
RE49083	12/2021	Case et al.	N/A	N/A
11339638	12/2021	Yeung et al.	N/A	N/A
11346200	12/2021	Cai et al.	N/A	N/A
11373058	12/2021	Jaaskelainen et al.	N/A	N/A
RE49140	12/2021	Case et al.	N/A	N/A

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RE49155	12/2021	Case et al.	N/A	N/A
RE49156	12/2021	Case et al.	N/A	N/A
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11441483	12/2021	Li et al.	N/A	N/A
11448122	12/2021	Feng et al.	N/A	N/A
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11492887	12/2021	Cui et al.	N/A	N/A
11499405	12/2021	Zhang et al.	N/A	N/A
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11512570	12/2021	Yeung	N/A	N/A
11519395	12/2021	Zhang et al.	N/A	N/A
11519405	12/2021	Deng et al.	N/A	N/A
11530602	12/2021	Yeung et al.	N/A	N/A
11549349	12/2022	Wang et al.	N/A	N/A
11555390	12/2022	Cui et al.	N/A	N/A
11555756	12/2022	Yeung et al.	N/A	N/A
11557887	12/2022	Ji et al.	N/A	N/A
11560779	12/2022	Mao et al.	N/A	N/A
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11575249	12/2022	Ji et al.	N/A	N/A
11592020	12/2022	Chang et al.	N/A	N/A
11596047	12/2022	Liu et al.	N/A	N/A
11598263	12/2022	Yeung et al.	N/A	N/A
11603797	12/2022	Zhang et al.	N/A	N/A
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11629589	12/2022	Lin et al.	N/A	N/A
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11668173	12/2022	Zhang et al.	N/A	N/A
11668289	12/2022	Chang et al.	N/A	N/A
11677238	12/2022	Liu et al.	N/A	N/A
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2002/0197176	12/2001	Kondo	N/A	N/A
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2005/0226754	12/2004	Orr et al.	N/A	N/A
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2006/0196251	12/2005	Richey	N/A	N/A
2006/0211356	12/2005	Grassman	N/A	N/A
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2006/0260331	12/2005	Andreychuk	N/A	N/A
2006/0272333	12/2005	Sundin	N/A	N/A
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2007/0107981	12/2006	Sicotte	N/A	N/A
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2007/0181212	12/2006	Fell	N/A	N/A
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2008/0161974	12/2007	Alston	N/A	N/A
2008/0212275	12/2007	Waryck et al.	N/A	N/A
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2008/0264649	12/2007	Crawford	N/A	N/A
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2010/0300683	12/2009	Looper et al.	N/A	N/A
2010/0310384	12/2009	Stephenson et al.	N/A	N/A
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2011/0052423	12/2010	Gambier et al.	N/A	N/A
2011/0054704	12/2010	Karpman et al.	N/A	N/A
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2011/0146244	12/2010	Farman et al.	N/A	N/A
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2011/0173991	12/2010	Dean	N/A	N/A
2011/0197988	12/2010	Van Vliet et al.	N/A	N/A
2011/0241888	12/2010	Lu et al.	N/A	N/A
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2011/0272158	12/2010	Neal	N/A	N/A
2012/0023973	12/2011	Mayorca	N/A	N/A
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2012/0255734	12/2011	Coli et al.	N/A	N/A
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2012/0324903	12/2011	Dewis et al.	N/A	N/A
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2013/0087045	12/2012	Sullivan et al.	N/A	N/A
2013/0087945	12/2012	Kusters et al.	N/A	N/A
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2013/0189915	12/2012	Hazard	N/A	N/A
2013/0205798	12/2012	Kwok et al.	N/A	N/A
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2013/0255953	12/2012	Tudor	N/A	N/A
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2014/0158345	12/2013	Jang et al.	N/A	N/A
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2015/0217672	12/2014	Shampine et al.	N/A	N/A
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2043184	12/1993	CA	N/A
2829762	12/2011	CA	N/A
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Background/Summary

PRIORITY CLAIM (1) This is a continuation of U.S. Non-Provisional application Ser. No. 17/383,517, filed Jul. 23, 2021, titled “COVER FOR FLUID SYSTEMS AND RELATED METHODS,” now U.S. Pat. No. 11,635,074, issued Apr. 25, 2023, which is a continuation of U.S. Non-Provisional application Ser. No. 15/929,652, filed May 14, 2020, titled “COVER FOR FLUID SYSTEMS AND RELATED METHODS,” which claims priority to and the benefit of, U.S. Provisional Application No. 62/704,462, filed May 12, 2020, titled “COVER FOR FLUID SYSTEMS AND RELATED METHODS,” and U.S. Provisional Application No. 62/704,476, filed May 12, 2020, titled “COVER FOR FLUID SYSTEMS AND RELATED METHODS,” the disclosures of which are incorporated herein by reference in their entireties.

BACKGROUND
(1) The present disclosure generally relates to pressurized fluid systems, fluid end covers, and methods, for example, high pressure single acting reciprocating pumping systems and methods such as those that include hydraulic fracturing single acting reciprocating pumps. Specifically, the present disclosure relates to a fluid cover for a fluid end block assembly of hydraulic fracturing pumping systems.
(2) Hydraulic fracturing often is used to produce oil and gas in an economic manner from low permeability reservoir rocks, such as shale. Hydraulic fracturing restores or enhances productivity of a well by creating a conductive flow path of hydrocarbons between the reservoir rock and a wellbore. During hydraulic fracturing, a fluid initially is pumped under high pressure to fracture rock in a reservoir formation and open a flow channel. Thereafter, a proppant-carrying fluid, e.g., a fluid that comprises proppant in the form of granular solid and/or semi-solid components, e.g., sand, ceramics, is pumped to continue opening and widening the flow channel while suspending proppant inside it. The proppant thus keeps the flow path opened for the hydrocarbons to flow.
(3) Hydraulic fracturing treatments may be performed using single acting reciprocating fracturing pumps to deliver fluids at a high pressure, specifically, above the fracture pressure of the rock in a reservoir formation. These fracturing pumps are of a type referred to as reciprocating plunger pumps. Such pumps may have multiple pumps, for example, 3 or 5 plungers to form “triplex” and “quintuplex” pumps, respectively. In such pumps, one or more plungers moves linearly back and forth in a cylindrical bore, traveling in and out of a pump fluid chamber. The fluid chamber is in communication with a suction or intake port and discharge port. Each port may include additional fluid handling components, for example, springs and valves (such as a one-way valve).
(4) In this regard, fluid enters the chamber through the suction port as the plunger withdraws from the chamber. It is then pumped out of the chamber through the discharge port as the plunger enters the chamber.
(5) The plungers are part of what is generally referred to as the fluid end of the pumping system, a major component of which is a pump

housing or fluid end block assembly. Accordingly, the fluid end block assembly may include passages or bores, e.g., cylindrical bores, in which the plungers travel, and within which valves, suction and discharge ports, fluid covers and other closures, etc., may be positioned.
(6) The cyclic movement of the plungers may forcibly pressurize fluids inside the fluid end block assembly of the pumping system. The rugged environments, high pressure of the fluid, and high power operations of the pumps causes fluid ends to become damaged, broken, and unusable, may decrease the usable life of the fluid ends, and may cause operational downtime or increased costs associated with pumping system operations at a well site.
SUMMARY
(7) In view of the foregoing, there is an ongoing need for enhanced pumping system components and methods more suitable for use in the fluid end block assembly of a pumping system when being used in the associated rugged environments, as well as for high pressure and high power operations.
(8) According to one embodiment of the disclosure, a high-pressure, high power, reciprocating positive displacement fluid pumping system may include a high-pressure, high power, reciprocating positive displacement pump having a pump plunger, a fluid end block assembly, and a fluid cover. The fluid end block assembly, for example, may include a fluid end block body, a suction port, a discharge port, a pump bore positioned in and extending through the fluid end block body, and a fluid chamber positioned in the fluid end block body and in fluid communication with each of the suction port, the discharge port, and the pump bore. The fluid chamber has an open end portion, and the pump plunger is positioned to move in the pump bore to pressurize one or more fluids in the fluid chamber. An embodiment of the fluid cover, for example, may have a monolithic body including a first portion and a second portion. The first portion may be received in the open end portion and sealably be engaged with the fluid end block body. The second portion may be connected mechanically to the fluid end block body. The fluid cover may include a shoulder that is positioned between the first portion and the second portion. The shoulder may be engaged with the fluid block end body to seal the open end portion. In some embodiments, the shoulder may interact with the open end portion to align the fluid cover with the open end portion.
(9) According to another embodiment of the disclosure, a fluid end block assembly for a high-pressure, high power, reciprocating positive displacement fluid pumping system may include a fluid end block body, a pump bore positioned in and extending through the fluid end block body, a plurality of ports including a suction port and a discharge port which collectively provide access to and from the pump bore, and a fluid chamber positioned in the fluid end block body. The fluid chamber may be positioned in fluid communication with each of the suction port, the discharge port, and the pump bore. The fluid chamber, for example, has an open end portion, and the pump bore may be positioned to receive a movable pump plunger to pressurize one or more fluids when located in the fluid chamber. A fluid cover may be connected to the fluid end block body, and the fluid cover may include a monolithic body having a first portion and a second portion. The first portion may be received in the open end portion and may be configured to align the fluid cover with the open end portion. The second portion may be connected mechanically to the fluid end block body.
(10) According to yet another embodiment of the disclosure, a fluid cover for being sealably engaged with a fluid end block assembly of a high-pressure, high power, reciprocating positive displacement fluid pumping system may include a monolithic body with a first portion and a second portion. The first portion, for example, may be received in an open end portion of a fluid chamber of the fluid end block assembly and sealably may be engaged with the fluid end block assembly. The first portion may be configured to engage the open end portion of the fluid chamber to align the fluid cover with the open end portion. The second portion may be connected mechanically to the fluid end block assembly.
(11) An embodiment of the disclosure also provides a method of operating a high-pressure, high power, reciprocating positive displacement fluid pumping system that includes obtaining a fluid end block assembly that has a fluid end block body, suction port, a discharge port, a pump bore positioned in and extending through the fluid end block body, and a fluid chamber positioned in the fluid end block body and in fluid communication with each of the suction port, the discharge port, and the pump bore. The fluid chamber may have an open end portion. The method further may include obtaining a high-pressure, high power, reciprocating positive displacement pump that has a pump plunger and fluidly connecting the pump to the fluid end block assembly. The method may include aligning a fluid cover with the fluid end block by inserting a first portion of the fluid cover into the open end portion. The first portion may engage the open end portion to align the fluid cover with the fluid end block. The fluid cover may include a monolithic body including the first portion and a second portion. The method further includes sealably connecting the fluid cover to the fluid end block by mechanically connecting the second portion to the fluid end block body. The method further may include operating the pump. The operating of the pump includes moving the pump plunger in the pump bore to pressurize one or more fluids in the fluid chamber.
(12) An embodiment of the disclosure also provides a method of assembly a fluid pumping system that includes inserting a first portion of a fluid cover into an open end portion of a fluid end block assembly and mechanically connecting a second portion of the fluid cover to the open end portion. The first portion may engage the fluid end block assembly to align the fluid cover with the open end portion. The fluid cover may comprise a monolithic body having the first and second portion. Mechanically connecting the second portion of the fluid cover may include the first portion sealing the open end portion of the fluid end block.
(13) Those skilled in the art will appreciate the benefits of various additional embodiments reading the following detailed description of the embodiments with reference to the below-listed drawing figures. It is within the scope of the present disclosure that the above-discussed aspects be provided both individually and in various combinations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) According to common practice, the various features of the drawings discussed below are not necessarily drawn to scale. Dimensions of various features and elements in the drawings may be expanded or reduced to more clearly illustrate the embodiments of the disclosure.
(2) FIG. **1**A is a schematic diagram of showing a layout of a fluid pumping system according to an embodiment of the disclosure.
(3) FIG. **1**B is an enlarged, sectional view of a fluid end block assembly of the fluid pumping system of FIG. **1**A according to an embodiment of the disclosure.
(4) FIG. **2**A is a perspective view of a fluid cover component of a conventional fluid cover assembly according to an embodiment of the disclosure.
(5) FIG. **2**B is a side view of the fluid cover component of FIG. **2**A according to an embodiment of the disclosure.
(6) FIG. **3**A is a perspective view of a retainer component of a conventional fluid cover assembly according to an embodiment of the disclosure.
(7) FIG. **3**B is a side view of the retainer component of FIG. **3**A according to an embodiment of the disclosure.
(8) FIG. **4** is an enlarged, sectional view of an operation of the fluid end block assembly of FIG. **1**B according to an embodiment of the

disclosure.
(9) FIG. **5** is another enlarged, sectional view of the fluid end block assembly of FIG. **1**B according to an embodiment of the disclosure.
(10) FIG. **6**A is a perspective view of an integrated fluid cover according to a first exemplary embodiment of the disclosure.
(11) FIG. **6**B is a side view of the integrated fluid cover of FIG. **6**A according to an embodiment of the disclosure.
(12) FIG. **6**C is a top view of the integrated fluid cover of FIG. **6**A according to an embodiment of the disclosure.
(13) FIG. **7**A is a side view of an integrated fluid cover according to a second exemplary embodiment of the disclosure.
(14) FIG. **7**B is another side view of the integrated fluid cover of FIG. **7**A, with a seal member shown in cross-section according to an embodiment of the disclosure.
(15) FIG. **8** is a sectional view of the integrated fluid cover of FIG. **7**A sealably coupled with a fluid end block assembly of a fluid pumping system according to an embodiment of the disclosure.
(16) FIG. **9** is a side view of an integrated fluid cover according to a third exemplary embodiment of the disclosure.
(17) FIG. **10** is a sectional view of the integrated fluid cover of FIG. **9** sealably coupled with a fluid end block assembly of a fluid pumping system according to an embodiment of the disclosure.
(18) Corresponding parts are designated by corresponding reference numbers throughout the drawings.

DETAILED DESCRIPTION

(19) The embodiments of the present disclosure are directed to pressurized fluid systems, for example, high pressure single acting reciprocating pumping systems such as those that include hydraulic fracturing single acting reciprocating pumps. In particular, the embodiments of the present disclosure are directed to fluid covers for use with such pressurized fluid systems.
(20) FIG. **1**A illustrates a schematic view of a high-pressure, high power, reciprocating positive displacement fluid pumping system **100** according to an embodiment of the disclosure. The diagram of the system **100** shows a typical pad layout for a fracturing pump system, and the system **100** includes a plurality of fracturing or frac pumps or pumping units FP**1**, FP**2**, FP**3**, FP**4**, FP**5**, FP**6**, FP**7**, FP**8** with the pumping units all operatively connected to a manifold M that is operatively connected to a wellhead W. By way of an example, in order to achieve a maximum rated horsepower of 24,000 HP for the pumping system **100**, a quantity of eight (8) 3000 horsepower (HP) pumping units may be used. It will be understood that the fluid pumping system **100** may include associated service equipment such as hoses, connections, and assemblies, among other devices and tools. Each of the pumping units FP**1**, FP**2**, FP**3**, FP**4**, FP**5**, FP**6**, FP**7**, FP**8** may include a fluid end block assembly **101**.
(21) FIG. **1**B illustrates a portion of a fluid end block assembly **101** of the embodiment of the fluid pumping system **100** as shown in a sectional view. In one embodiment, the fluid end block assembly **101** may be in fluid communication with and/or may form a portion of the manifold M. In this regard, one or more of the frac pumps or pumping units FP**1**, FP**2**, FP**3**, FP**4**, FP**5**, FP**6**, FP**7**, FP**8** may be fluidly coupled to the fluid end block assembly **101**.
(22) The fluid end block assembly **101**, as shown, includes a fluid end block body **121** having intersecting horizontal and vertical bore passages that are each in fluid communication with a fluid end chamber **105** in the fluid end block body **121**. The vertical bore passage includes an intake port or suction port **102**. The fluid may enter the fluid end block body **121** from an intake source, for example, a fluid supply manifold. The vertical bore passage also includes an outlet port or discharge port **103** through which fluid may exit the fluid end block body **121** or flow to another portion of the fluid end block assembly **101**. One or both of the suction port **102** and the discharge port **103** thus collectively provide access to and from the fluid end chamber **105** and may include fluid handling elements or components such as seats, valves, springs, and so forth, as will be understood by those skilled in the art.
(23) The discharge port **103**, as shown, includes a discharge bore **104**, e.g., an opening or fluid channel, through which pressurized fluid may exit the fluid end block assembly **101** to downstream components of the fluid pumping system coupled to the fluid end block assembly **101**. A discharge cover assembly or discharge cap assembly **116** may be coupled to a portion of the discharge port **103**, as described further herein.
(24) As shown, a piston/plunger **107** (broadly, “pump member”) is positioned in a pump bore **108** along the horizontal bore of the fluid end block assembly **101**. The plunger **107** is movable in the pump bore **108**, for example, via reciprocating actuation of one or more of the pumping units FP**1**, FP**2**, FP**3**, FP**4**, FP**5**, FP**6**, FP**7**, FP**8**, with a forward stroke in the direction of the fluid chamber **105** and that terminates proximate the fluid chamber **105**, and a rearward stroke in a direction away from the fluid chamber **105** and opposite the direction of the forward stroke. One or more of the pumping units FP**1**, FP**2**, FP**3**, FP**4**, FP**5**, FP**6**, FP**7**, FP**8** may actuate the plunger **107** to move/reciprocate in the pump bore **108**, for example, via a controller or control system, as will be understood by those skilled in the art, that may be in electronic communication with one or more of the pumping units, and which may be operated under manual and/or processor control.
(25) As shown, the forward stroke of the plunger **107** terminates opposite an open end portion **106** of the fluid chamber **105**. In one embodiment, the open end portion **106** of the fluid chamber **105** is positioned in a portion of the fluid end block body **121** that includes an oblique or chamfered surface **123** that extends toward an interior threaded portion **127** of the fluid end block body **121** along interior peripheries of the open end portion **106** of the fluid chamber **105** (see FIG. **5**).
(26) A fluid cover assembly **109** (broadly, “suction cap assembly” or “suction cover assembly” or “intake cap assembly” or “intake cover assembly”) may be sealingly engaged with and coupled to the fluid end block body **121**, as described further herein.
(27) An embodiment of the fluid cover assembly **109** may include a fluid cover component **110** having a generally cylindrical body with a flanged head **111** including a tool engagement feature **113** for being engaged by a tool to place, position, help secure to, and/or remove the fluid cover component **110** from the end portion **106** of the fluid chamber **105** (see FIGS. **2**A-**2**B and **3**A-**3**B). In this regard, the fluid cover component **110** may be at least partially received within the end portion **106** of the fluid chamber **105** opposite the reciprocating path of the plunger **107**. A sealing member **114**, e.g., a rubber or other polymeric annular seal, may be positioned on the cylindrical body of the fluid cover component **110**.
(28) The fluid cover component **110** may be at least partially received in the end portion **106** of the fluid chamber **105**, and the fluid cover assembly **109** may be mechanically connected to the fluid end block assembly **101** via the mechanical connection of a retainer component **112** to the fluid end block body **121**.
(29) The retainer component **112**, as shown in FIGS. **3**A and **3**B, may be a retainer nut or other fastener having a generally cylindrical body with an outer surface that defines a threaded portion **126** that is configured to threadably engage the corresponding interior threaded portion **127** of the fluid end block body **121**. In this regard, the retainer component **112** may be threadably advanced into the end portion **106** of the fluid chamber **105** to abut the flanged head portion **111** of the fluid cover component **110**, for example, such that an end of the retainer component **112** contacts the head portion of the fluid cover component **112** at an interface, e.g., a discontinuity between the material of the fluid cover component **110** and the material of the retainer component **112**. As shown, the retainer component **112** may also define a tool engagement feature **117** for being engaged by a tool to place, position, and/or remove the retainer component **112** from the end portion **106** of the fluid chamber **105**.
(30) It will be understood that the discharge cap/cover assembly **116** of the discharge port **103** may have a configuration that is generally

similar to the configuration of the fluid cover assembly **109** positioned at the open end portion **106** of the fluid chamber **105** near the suction port **102**. The discharge cap/cover assembly **116** of the discharge port **103** may have a different configuration without departing from the disclosure.

(31) Referring to FIG. **4**, fluid may be delivered via the suction port **102**, for example, from a pump suction manifold and valve assembly, into the fluid chamber **105**. The fluid in the fluid chamber **105** may be compressed by the plunger **107** upon a compression or delivery stroke, e.g., toward the fluid chamber **105**.

(32) In the embodiment illustrated in FIG. **4**, the full extension length of the plunger **107** may extend at least partially toward the fluid chamber **105** to compress fluids disposed therein. In this regard, as the plunger **107** commences its delivery or compression stroke, the fluid positioned inside of the fluid end chamber **105** is compressed by the plunger **107** and such compression forces are translated into the surrounding walls of the fluid chamber **105**. As the end portion **106** of the fluid chamber **105** is closed by the fluid cover assembly **109** sealingly engaged with the fluid end block body **121**, the pressurized fluid in the fluid end chamber **105** flows into the discharge port **103**, and out the discharge bore **104**.

(33) Following/preceding the above-described compression or delivery stroke of the plunger **107**, in an intake or suction stroke, the plunger **107** translates along the pump bore **108** of the fluid end block assembly **101** away from the fluid chamber **105** to create a suction, e.g., negative pressure or vacuum, in the fluid chamber **105** that draws additional fluid into the fluid chamber **105** in preparation for a subsequent compression or delivery stroke.

(34) Referring to FIGS. **6**A-**6**C, an integrated fluid cover according to a first exemplary embodiment of the disclosure is generally designated **221**, and may be used in a fluid pumping system that may be otherwise similar to the fluid pumping system **100** described above.

(35) In the illustrated embodiment, the integrated fluid cover **221** includes a monolithic body **223**, e.g., a body monolithically formed of a single continuous piece or block of material that has a unitary configuration so as to be free from any seams or discontinuities that extend from an outer surface of the body **223** to an interior portion of the body **223**. It will be understood that the body **223** of the integrated fluid cover **221** may include metallic, polymeric, and/or composite materials.

(36) The body **223** of the integrated fluid cover **221** defines a fluid cover portion **225** (broadly, “first portion”) and a retainer portion **227** (broadly, “second portion”). Each of the fluid cover portion **225** and the retainer portion **227** has a generally cylindrical configuration, with the fluid cover portion **225** having a first diameter D**1** that is smaller than a second diameter D**2** of the fluid cover portion **225**.

(37) The fluid cover portion **225** has a generally cylindrical configuration with a free end or distal end **226** facing the fluid chamber **105**. The fluid cover portion **225** extends from the distal end **226** to a flange portion **228** of the retainer portion **227**. In this regard, the fluid cover portion **225** of the integrated fluid cover **221** may be at least partially received within the end portion **106** of the fluid chamber **105** of the fluid end block assembly **101**.

(38) As shown, the flange portion **228** of the retainer portion **227** extends radially outwardly from the fluid cover portion **225**. The retainer portion **227** also includes a threaded portion **229** extending away from the flange portion **228** and that is configured to engage the interior threaded portion **127** of the fluid end block body **121** to mechanically connect to the fluid end block body **121** as described above. In one embodiment, the flange portion **228** of the retainer portion **227** may extend radially outwardly from the threaded portion **229** so as to form a protrusion or protuberance along the outer surface of the retainer portion **227**.

(39) As shown, the seal member **114** (broadly, “first seal member”) may be positioned on the fluid cover portion **225** of the integrated fluid cover **221** between the distal end **226** and the flange portion **228** of the retainer portion **227**. In this regard, when the fluid cover portion **225** of the integrated fluid cover **221** is at least partially received in the end portion **106** of the fluid chamber **105**, the seal member **114** may be positioned to sealingly engage interior surfaces of the fluid end block body **121**. The seal member **114** may align or center the fluid cover portion **225** within the end portion **106** of the fluid chamber **105**.

(40) As also shown, the body **223** of the integrated fluid cover **221** defines a tool engagement feature **231** at a free end or proximal end surface **232** of the retainer portion **227**. The tool engagement feature **231** may be, for example and without limitation, a polygonal (e.g., hexagonal) recess having a configuration complementary to that of an insertion or removal or securing tool, such as an Allen wrench or other driver when positioned to secure the integrated fluid cover **221**. Optionally, the tool engagement feature may have a different configuration, for example and without limitation, a protrusion, without departing from the disclosure.

(41) In this regard, rotation of the integrated fluid cover **221** via engagement of a tool with the tool engagement feature **231** may cause rotation of both the fluid cover portion **225** and the retainer portion **227** of the integrated fluid cover **221** due to the monolithic construction of the body **223** of the integrated fluid cover **221**. Such a configuration may obviate and reduce the number of tools required for insertion, removal, and other maintenance of a fluid cover in which, for example, a retainer portion and a fluid cover portion are separate components.

(42) Furthermore, the aforementioned monolithic construction of the body **223** of the integrated fluid cover **221** may provide a higher mass single body as compared to the bodies of a fluid cover portion and a retainer portion provided as separate components of a fluid cover assembly. In this regard, the integrated fluid cover **221** provides enhanced material integrity, durability, and fatigue resistance in high pressure fluid environments.

(43) For example, in a conventional fluid cover assembly that includes a separately coupled fluid cover portion and retainer portion, cyclic high fluid pressures produced in a fluid pump system may result in a water hammer effect in which impacts the fluid cover portion and which is translated towards the retainer portion. These fluid pulsations may be influenced by factors such as pump operating pressure, pump crankshaft rotation speed, suction and discharge valve efficiency, and effective fluid end chamber fill volume per plunger stroke. The fluid pulses/forces thus produce a constant vibration and wear on components of the fluid block assembly **101**, and may cause relative movement of the fluid cover portion and the retainer portion, which may result, for example, in wearing down or away of threaded portions or other coupling features that may create a clearance gap (see FIG. **5**) between the fluid cover portion and the separate retainer portion, backing out or disengagement of the retainer portion from an end portion of a fluid end block, the loss of seal integrity between the fluid cover portion and the end portion of the fluid end block that may lead to leakage of fluids from the fluid end block, etc.

(44) Accordingly, the disclosed integrated fluid cover **221** is resistant to wear and failure produced in the cyclic high pressure fluid environments in a fluid end block assembly **101** so as to reduce damage (e.g., wash, wear, cracking, etc.), reduce maintenance cycles and downtimes, minimize consumable components for the fluid end block assembly **101**, ensure the maintenance of proper sealing contact with surfaces of the fluid end block assembly **101**, and reduce leakage from the fluid end block assembly **101**.

(45) In addition, because the tool engagement feature **231** may be engaged to both position and seat the fluid cover portion **225** in the end portion **106** of the fluid end block assembly **101** as well as engage/couple the retainer portion **227** with the fluid end block body **121**, the number of tools employed for installation/maintenance of components of the fluid end block assembly **101** may be reduced, as well as obviating the need to align a separate fluid cover portion and a retainer portion.

(46) Further, constructing or manufacturing the fluid cover portion **225** and the retainer portion **227** into a single monolithic body of the fluid cover **221** may increase resistance to backing out and/or rotation of the fluid cover **221** during the rugged environment associated with

operation of a fluid end. For example, with a separate fluid cover portion and retainer portion (see FIGS. **2**A-**3**B), the fluid cover portion is subject to translation forces towards and away from the retainer portion as a result of what is known as the water hammer effect as will be understood by those skilled in the art, and the retainer portion is subject to rotational forces as a result of the retainer portion engaging the flange thereof. In contrast, the monolithic body of the fluid cover **221** increases a mass resisting the water hammer effect. In addition, as the retainer portion **227** and the fluid cover portion **225** are monolithically formed together, the water hammer effect may be reduced by eliminating movement between the retainer portion **227** and the fluid cover portion **225**. Additionally, the frictional engagement between the monolithic fluid cover **221** may be increased to further resist the water hammer effect. For example, the seal member **114** frictionally engages the inner surface of the fluid end block body **121** which may increase resistance to rotation of the fluid cover portion **225** relative to the fluid end block body **121** when compared to a non-monolithically formed fluid cover. This engagement further may increase a service life of the fluid cover **221** and/or the fluid end block body **121** by reducing the number of component parts and by reducing the risk of failure of this particular component part due to its construction as illustrated and described in the associated embodiments.
(47) Referring to FIGS. **7**A and **7**B, an integrated fluid cover according to a second exemplary embodiment of the disclosure is generally designated **321**, and may be sealably coupled to the fluid end block assembly **101** of a fluid pumping system **300** that may be otherwise similar to the fluid pumping systems described above, e.g., system **100**. The integrated fluid cover **321** may have one or more features that are substantially similar to those described above with regard to the integrated fluid cover **221**, and like or similar features are designated with like or similar reference numerals.
(48) The integrated fluid cover **321** has a body **323** that is substantially similar to the body **223** of the integrated fluid cover **221**, except that the body **321** defines a machined or molded annular recess or groove **322** along the fluid cover portion **225** adjacent the flange portion **228** of the retainer portion **227** and for at least partially receiving a seal member **324** (broadly, “second seal member”) therein. The seal member **324** may be a flexible and/or resilient member, such as a polymeric (e.g., rubber) ring, for example and without limitation, an O-ring.
(49) In this regard, the annular groove **322** may provide for an arrangement of the seal member **324** about the fluid cover portion **225** of the integrated fluid cover **321** that has a low or minimized profile, e.g., such that a minimal portion of the seal member **324** extends above the annular groove **322**. Accordingly, the annular groove **322** and seal member **324** cooperate to provide an arrangement of the seal member **324** that enhances a fluid-resistant seal against the fluid end block body **121**, but does not interfere with proper placement or receipt of the fluid cover portion **225** in the end portion **106** of the fluid chamber **105**.
(50) Furthermore, and with additional reference to FIG. **8**, when the fluid cover portion **225** of the integrated fluid cover **321** is at least partially received with the end portion **106** of the fluid chamber **105**, the seal member **324** may be compressed to a reduced profile, which may also have the effect of enhancing the sealing engagement of the seal member **324** against the fluid end block body **121**. As shown in FIG. **8**, the flange portion **228** of the integrated fluid cover **321** is positioned to seat against an internal surface **128** of end portion **106** of the fluid chamber **105**. Specifically, the seal member **324** may be configured to engage the chamfered surface **123** of the fluid end block body **121** such that the seal member **324** fills space or at least some of the interstitial space between the retainer portion **225** and the fluid end block body **121** as will be understood by those skilled in the art.
(51) Referring now to FIG. **9**, an integrated fluid cover according to a third exemplary embodiment of the disclosure is generally designated **421**, which may be sealably coupled to the fluid end block assembly **101** of a fluid pumping system **400** that may be otherwise similar to the fluid pumping systems described above, e.g., systems **100**, **300**. The integrated fluid cover **421** may have one or more features that are substantially similar to those described above with regard to the integrated fluid covers **221**, **321** and like or similar features are designated with like or similar reference numerals.
(52) The integrated fluid cover **421** has a body **423** that is substantially similar to the body **223** of the integrated fluid cover **221**, except that the body **421** includes a neck portion or shoulder **422** extending from a surface of the fluid cover portion **225** to the flange portion **228** of the retainer portion **227**. In this regard, the shoulder **422** of the integrated fluid cover **421** has a tapered configuration that provides an oblique annular surface positioned between the fluid cover portion **225** and the retainer portion **227** to substantially surround the fluid cover **421**. The oblique annular surface of the shoulder **422** increases in diameter as the shoulder **422** extends towards the flange portion **228** of the retainer portion **227**.
(53) Accordingly, and with additional reference to FIG. **10**, when the fluid cover portion **225** of the integrated fluid cover **421** is at least partially received with the end portion **106** of the fluid chamber **105**, the shoulder **422** may sealingly engage the chamfered surface **123** in a frictional and/or wedged arrangement so as to provide an enhanced seal of the integrated fluid cover **421** against the fluid end block body **121** that may resist fluid leakage from the fluid chamber **105**. In one embodiment, both the fluid end block body **101** and at least the shoulder **422** of the integrated fluid cover **421** may be comprised of metal such that the interface of the shoulder **422** with the outer edge of the end portion **106** of the fluid chamber is a metal-to-metal interface that provides an enhanced seal. In some embodiments, the shoulder **422** may interact with the chamfered surface **123** to align or center the integrated fluid cover **421** with the end portion **106** of the fluid chamber **105**.
(54) This is a continuation of U.S. Non-Provisional application Ser. No. 17/383,517, filed Jul. 23, 2021, titled “COVER FOR FLUID SYSTEMS AND RELATED METHODS,” now U.S. Pat. No. 11,635,074, issued Apr. 25, 2023, which is a continuation of U.S. Non-Provisional application Ser. No. 15/929,652, filed May 14, 2020, titled “COVER FOR FLUID SYSTEMS AND RELATED METHODS,” which claims priority to and the benefit of, U.S. Provisional Application No. 62/704,462, filed May 12, 2020, titled “COVER FOR FLUID SYSTEMS AND RELATED METHODS,” and U.S. Provisional Application No. 62/704,476, filed May 12, 2020, titled “COVER FOR FLUID SYSTEMS AND RELATED METHODS,” the disclosures of which are incorporated herein by reference in their entireties.
(55) The foregoing description of the disclosure illustrates and describes various exemplary embodiments. Various additions, modifications, changes, etc., could be made to the exemplary embodiments without departing from the spirit and scope of the disclosure. It is intended that all matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense. Additionally, the disclosure shows and describes only selected embodiments of the disclosure, but the disclosure is capable of use in various other combinations, modifications, and environments and is capable of changes or modifications within the scope of the inventive concept as expressed herein, commensurate with the above teachings, and/or within the skill or knowledge of the relevant art. Furthermore, certain features and characteristics of each embodiment may be selectively interchanged and applied to other illustrated and non-illustrated embodiments of the disclosure.

Claims

1. A method comprising: inserting a first portion of a fluid cover into an open end portion of a fluid end block assembly of a fluid pumping system, the fluid cover having a monolithic body including the first portion and a second portion such that the first portion extends from the second portion to an end of the monolithic body, the first portion including an annular groove disposed adjacent to the second portion; sealingly engaging at least one first seal member positioned on the first portion with an inner wall of the open end portion such that the first

portion is spaced away from the inner wall by the at least one first seal member; sealingly engaging at least one second seal member, which is seated at least partially in the annular groove, with an interior surface of the open end portion; and mechanically connecting the second portion of the fluid cover to the open end portion of the fluid end block assembly.
2. The method of claim 1, wherein the mechanically connecting the second portion comprises threadably engaging the second portion with the open end portion.
3. The method of claim 2, further comprising engaging a shoulder positioned within the open end portion with a flange defined on the second portion of the fluid cover.
4. The method of claim 3, wherein the second seal member is sealingly engaged against an interior chamfered surface of the open end portion that is positioned between the shoulder and the inner wall.
5. The method of claim 3, further comprising engaging an interior chamfered surface of the open end portion with a tapered shoulder of the fluid cover that is positioned adjacent the flange.
6. The method of claim 5, wherein the tapered shoulder has an increasing diameter as the tapered shoulder extends toward the flange along the fluid cover.
7. The method of claim 5, wherein the engaging the interior chamfered surface with the tapered shoulder comprises engaging the tapered shoulder with the interior chamfered surface in a wedged arrangement.
8. The method of claim 5, wherein the engaging the interior chamfered surface with the tapered shoulder comprises forming a metal-to-metal interface therebetween.
9. The method of claim 1, wherein the at least one first seal member and the at least one second seal member are each a polymeric O-ring.
10. A method comprising: obtaining a fluid end block assembly of a fluid pumping system, the fluid end block assembly including: a fluid chamber having an open end portion, the open end portion having an inner wall, and a fluid cover inserted into the open end portion, the fluid cover including a monolithic body having a first portion and a second portion, the first portion including an annular groove disposed adjacent to the second portion; sealingly engaging the inner wall with at least one first seal member that is positioned annularly about the first portion such that the first portion is spaced away from the inner wall by the at least one first seal member, thereby to align the fluid cover within the open end portion; sealingly engaging at least one second seal member, which is seated at least partially in the annular groove, with the inner wall; mechanically connecting the second portion to the open end portion; operating the fluid pumping system to pressurize one or more fluids in the fluid chamber; and preventing leakage of the one or more fluids from the fluid chamber through the open end portion with the at least one first seal member and the at least one second seal member during the operating.
11. The method of claim 10, wherein the mechanically connecting comprises threadably engaging the second portion with the open end portion.
12. The method of claim 11, further comprising engaging a shoulder positioned within the open end portion with a flange defined on the second portion of the fluid cover.
13. The method of claim 12, wherein the second seal member is sealingly engaged against an interior chamfered surface of the open end portion that is positioned between the shoulder and the inner wall.
14. The method of claim 12, further comprising engaging an interior chamfered surface of the open end portion with a tapered shoulder of the fluid cover that is positioned adjacent the flange.
15. The method of claim 14, wherein the tapered shoulder has an increasing diameter as the tapered shoulder extends toward the flange along the fluid cover.
16. The method of claim 14, wherein the engaging the interior chamfered surface with the tapered shoulder comprises engaging the tapered shoulder with the interior chamfered surface in a wedged arrangement to thereby further seal the open end portion.
17. The method of claim 10, wherein the at least one first seal member and the at least one second seal member are each a polymeric O-ring.
18. A method comprising: inserting a first portion of a fluid cover into an open end portion of a fluid end block assembly of a fluid pumping system, the fluid cover having a monolithic body including the first portion and a second portion such that the first portion extends from the second portion to an end of the monolithic body, the first portion including an annular groove; sealingly engaging at least one first seal member positioned on the first portion with an inner wall of the open end portion such that the first portion is spaced away from the inner wall by the at least one first seal member; threadably engaging the second portion of the fluid cover with the open end portion of the fluid end block assembly; engaging a shoulder positioned within the open end portion with a flange defined on the second portion of the fluid cover; and sealingly engaging a second seal member that is at least partially seated in the annular groove of the first portion with an interior chamfered surface of the open end portion that is positioned between the shoulder and the inner wall.
19. The method of claim 18, wherein the inner wall is a cylindrical inner wall, wherein the first portion has an outer cylindrical surface, and wherein the at least one first seal member is positioned annularly about the outer cylindrical surface.
20. The method of claim 18, wherein the at least one first seal member and the at least one second seal member are each a polymeric O-ring.